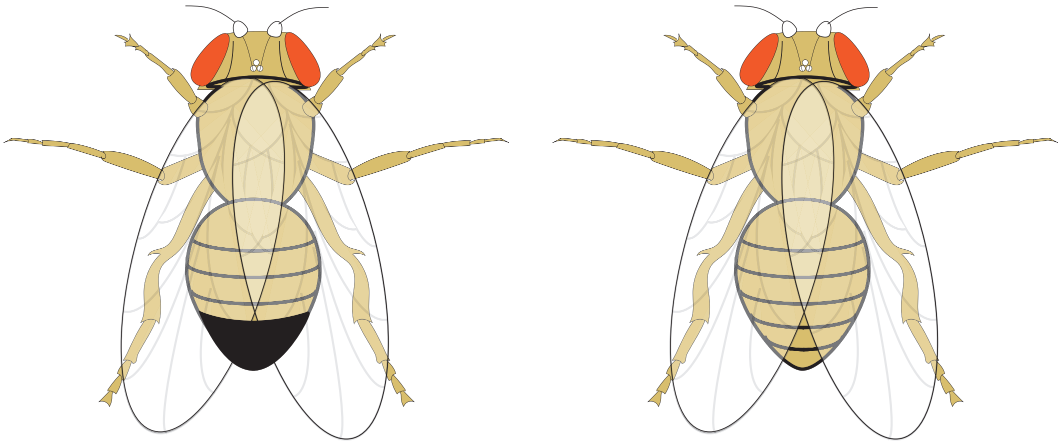
Examiner only

3. A class of year 11 students decided to carry out some genetics investigations on the fruit fly *Drosophila melanogaster*. Fruit flies are easily kept in school laboratories and are a good species for genetics investigations because they:

- have a short generation time (a very short time to develop from egg to adult and reproduce)
- are easy to maintain
- males (♂) and females (♀) are easy to identify
- they produce between 200 and 500 offspring per mating

The students were asked to draw a sketch of a male and a female fruit fly. A sketch is shown below.



♂ ♀

(a) Suggest an advantage to the students, of the fruit fly:

(i) having a short generation time; [1]

....

(ii) producing large numbers of offspring. [1]

....

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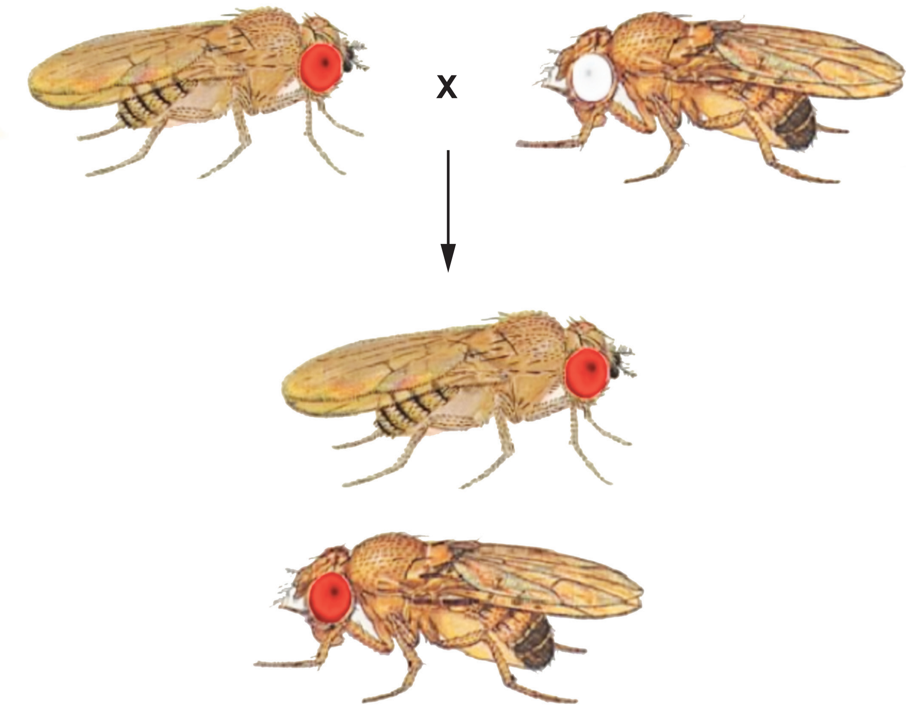
(b) The students were asked to cross a red-eyed fly and a white-eyed fly. One of each of these flies was placed in a specimen tube with some nutrients. The diagram below shows the phenotype of both the flies in this cross and some of the offspring.



All F1 offspring have red eyes

(i) **Label one** of the **male** flies in the diagram above. [1]

(ii) Show this cross in the Punnett square below. Use the letters **R** to represent the allele for red eye colour and the letter **r** to represent the alleles for white eye colour. [2]

F1

Gametes		

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(c) The students then decided to self the F1 generation. There were 8 groups of students. When they examined the F2 generation they obtained the following results.

Group number	Number of offspring in F2 generation	Number of flies with red eyes	Number of flies with white eyes	Approximate ratio of red-eyed to white-eyed flies
1	214	156	58	....
2	139	108	31	....
3	0	0	0	No ratio
4	276	206	70	....
5	319	244	75	....
6	39	28	11	....
7	217	55	162	1 : 3
8	312	235	77	....

(i) **Complete the table** by writing in the **approximate** ratio of red to white eyed flies for all the remaining groups. Use whole numbers only. [1]

(ii) The students in group number 3 followed the method correctly. Suggest a reason for their result. [1]

....

(iii) Suggest an explanation for the anomalous result obtained by group number 7. [1]

....

....

(iv) Complete the Punnett square below to show the expected result when the F1 were selfed. [2]

F2

Gametes		

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